ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

Kenan Song
School of Chips
Email:
Kenan.Song@xjtlu.edu.cn
Office: C5007

Real and Reciprocal Spaces II

I. The Fourier Transform of Matter

II. Reciprocal Space

III. The First Brillouin Zone (BZ)

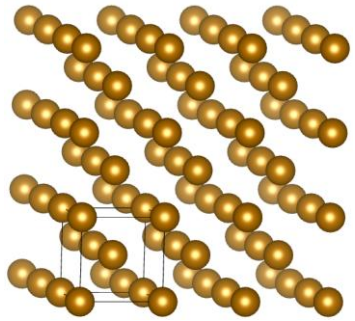
IV. High-Symmetry Points and Irreducible BZ

I. The Fourier Transform of Matter

An artistic overlay showing a real-space crystal lattice and its corresponding diffraction pattern (Reciprocal Space)

A diagram of *X*-ray diffraction (Bragg's Law) showing that the distance between diffraction spots is inversely proportional to the distance between atomic planes

In Real Space

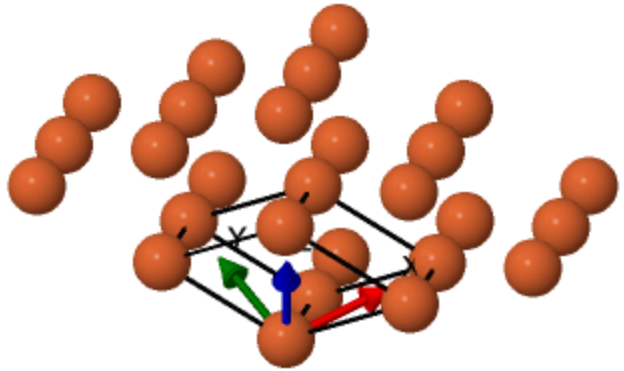


Atoms



Waves

In Reciprocal Space



Crystal Structure

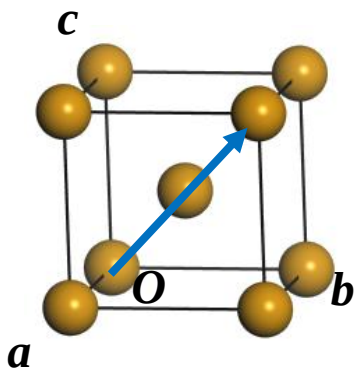
***k* Point Grid**

One see atoms (particles)

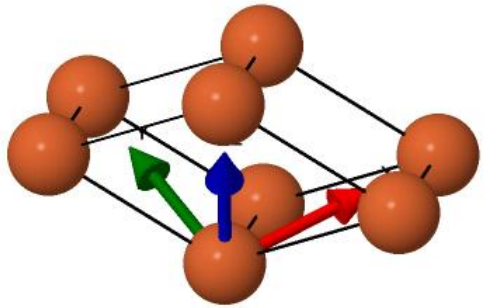
One sees momentum/waves (electrons/phonons)

I. The Fourier Transform of Matter

The Reciprocal Lattice is the Fourier Transform of the Real Lattice



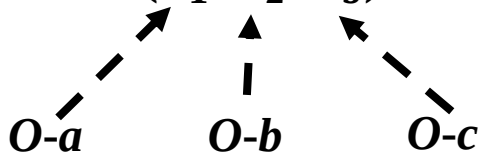
Fourier Transform



In Real Space

In Reciprocal Space

Lattice vectors (a_1, a_2, a_3) in Real Space



$$b_1 = 2\pi \frac{a_2 \times a_3}{V}$$

$$b_2 = 2\pi \frac{a_3 \times a_1}{V}$$

$$b_3 = 2\pi \frac{a_1 \times a_2}{V}$$

Lattice vectors (b_1, b_2, b_3) in Reciprocal Space

$$V = a_1 \cdot (a_2 \times a_3) = a_2 \cdot (a_3 \times a_1) = a_3 \cdot (a_1 \times a_2)$$

I. The Fourier Transform of Matter

In Real Space

$$\boldsymbol{a_1} = (a_{1x}, a_{1y}, a_{1z})$$

$$\boldsymbol{a_2} = (a_{2x}, a_{2y}, a_{2z})$$

$$\boldsymbol{a_3} = (a_{3x}, a_{3y}, a_{3z})$$

Inversely Proportional Relationship

$(a_1, a_2, a_3) \longleftrightarrow (b_1, b_2, b_3)$

In Reciprocal Space

$$\boldsymbol{b_1} = 2\pi \frac{\boldsymbol{a_2} \times \boldsymbol{a_3}}{V} = \frac{1}{\boldsymbol{a_1}}$$

$$\boldsymbol{b_2} = 2\pi \frac{\boldsymbol{a_3} \times \boldsymbol{a_1}}{V} = \frac{1}{\boldsymbol{a_2}}$$

$$\boldsymbol{b_3} = 2\pi \frac{\boldsymbol{a_1} \times \boldsymbol{a_2}}{V} = \frac{1}{\boldsymbol{a_3}}$$

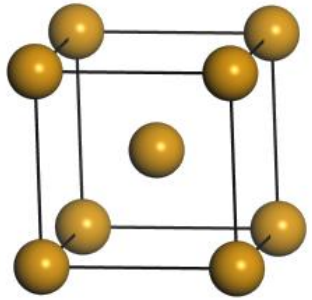
$$\boldsymbol{a_1} \times \boldsymbol{a_2} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ a_{1x} & a_{1y} & a_{1z} \\ a_{2x} & a_{2y} & a_{2z} \end{vmatrix} = (a_{1y} \times a_{2z} - a_{1z} \times a_{2y})\hat{x} + (a_{1z} \times a_{2x} - a_{1x} \times a_{2z})\hat{y} + (a_{1x} \times a_{2y} - a_{1y} \times a_{2x})\hat{z}$$

$$V = \boldsymbol{a_3} \cdot (\boldsymbol{a_1} \times \boldsymbol{a_2}) = a_{3x} \times (a_{1y} \times a_{2z} - a_{1z} \times a_{2y}) + a_{3y} \times (a_{1z} \times a_{2x} - a_{1x} \times a_{2z}) + a_{3z} \times (a_{1x} \times a_{2y} - a_{1y} \times a_{2x})$$

I. The Fourier Transform of Matter

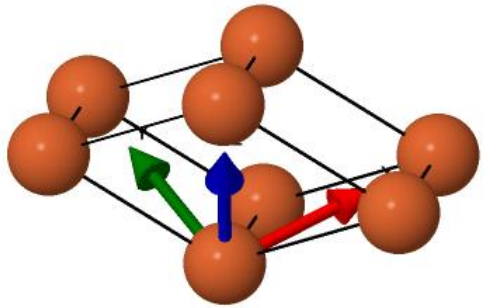
In Real Space

In Reciprocal Space



Fourier Transform

Vice Versa



Body Centered Cubic (BCC)

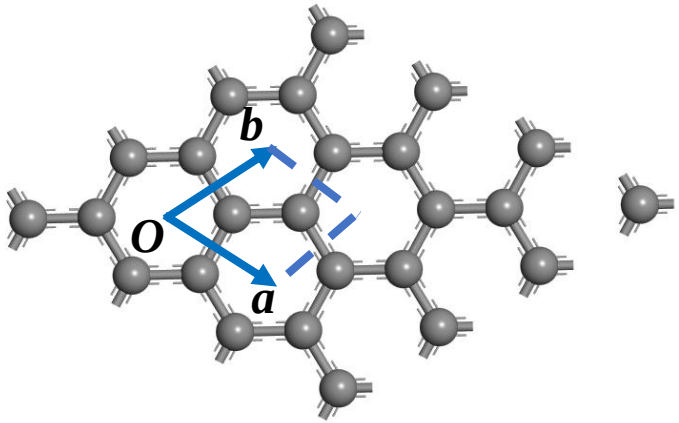
Face Centered Cubic (FCC)

When one calculates the band structure of Aluminum (FCC), calculation is actually performed within a BCC-shaped momentum space

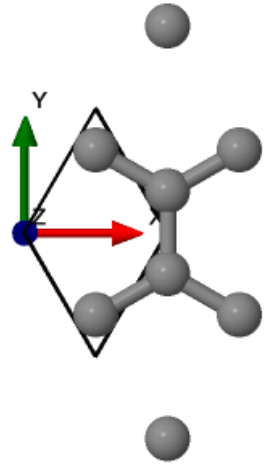
This briefly shows how the skewed primitive vectors of FCC transform into the primitive vectors of a BCC-like structure in the reciprocal space

I. The Fourier Transform of Matter

Special Case 2D System



Fourier Transform



In Real Space

In Reciprocal Space

Lattice vectors (a_1, a_2) in Real Space

$$a_1 = (a_{1x}, a_{1y}, 0)$$

$$a_2 = (a_{2x}, a_{2y}, 0)$$

$$\hat{n} = (0, 0, 1)$$

$$b_1 = 2\pi \frac{a_2 \times \hat{n}}{A}$$

$$b_2 = 2\pi \frac{a_2 \times \hat{n}}{A}$$

$$A = |a_1 \cdot (a_2 \times \hat{n})|$$

Lattice vectors (b_1, b_2) in Reciprocal Space

II. Reciprocal Space

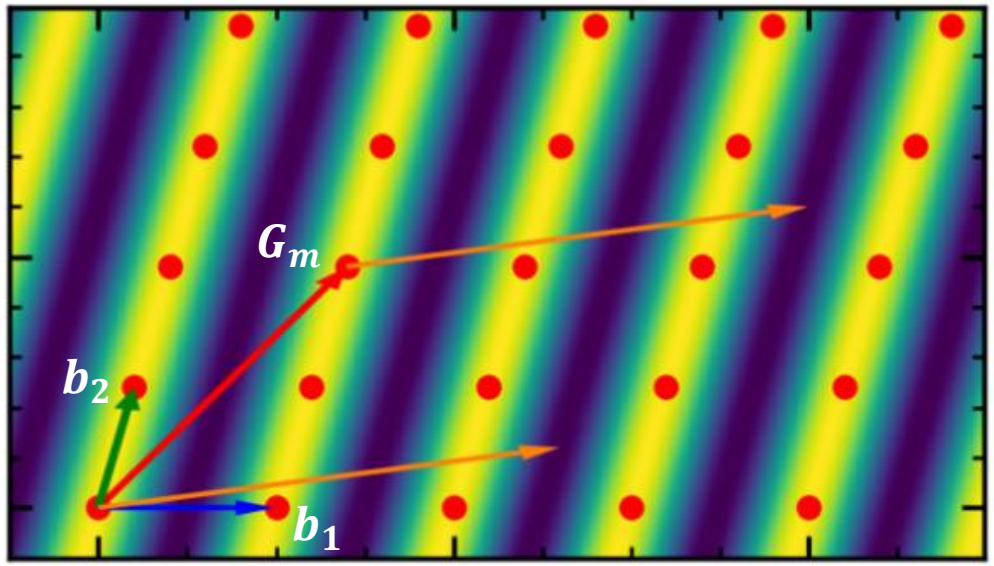
A concept associated with solids with translational symmetry which plays a major role in many areas such as *X*-ray and electron diffraction as well as the energies of electrons in a solid

It emerges from the Fourier transform of the lattice associated with the arrangement of the atoms

The direct lattice or real lattice is a periodic function in physical space, such as a crystal system (usually a Bravais lattice)

The reciprocal lattice exists in the mathematical space of spatial frequencies or wavenumbers *k*, known as reciprocal space or *k* space; it is the dual of physical space considered as a vector space

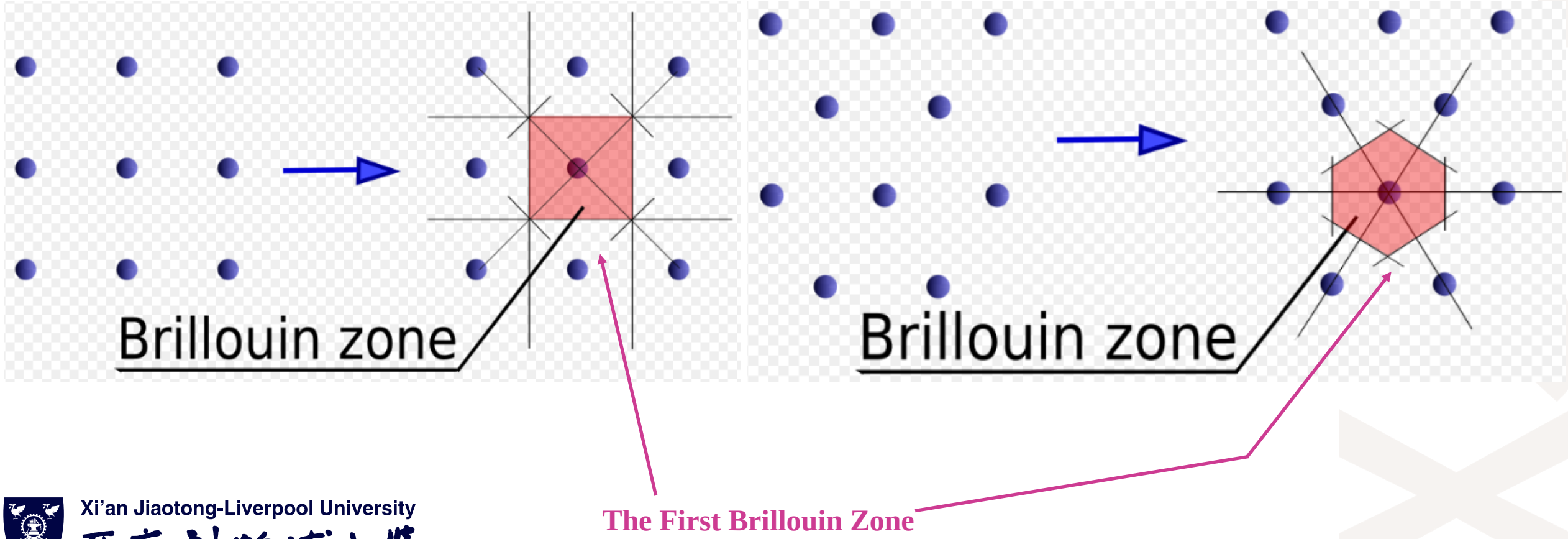
$$G_m = m_1 b_1 + m_2 b_2$$



III. The First Brillouin Zone (BZ)

The first Brillouin zone (named after Léon Brillouin) is a uniquely defined primitive cell in reciprocal space

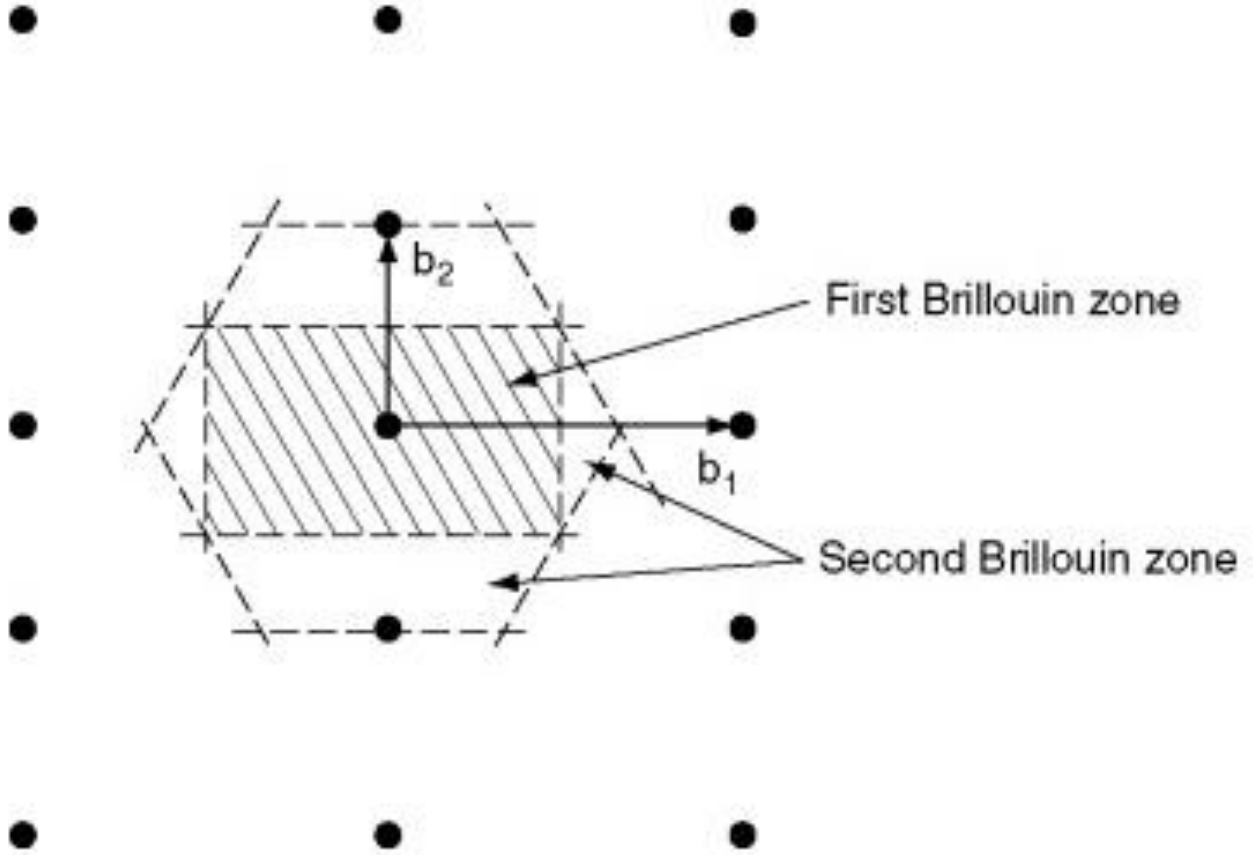
In the same way the Bravais lattice is divided up into Wigner–Seitz cells in the real lattice, the reciprocal lattice is broken up into Brillouin zones



III. The First Brillouin Zone (BZ)

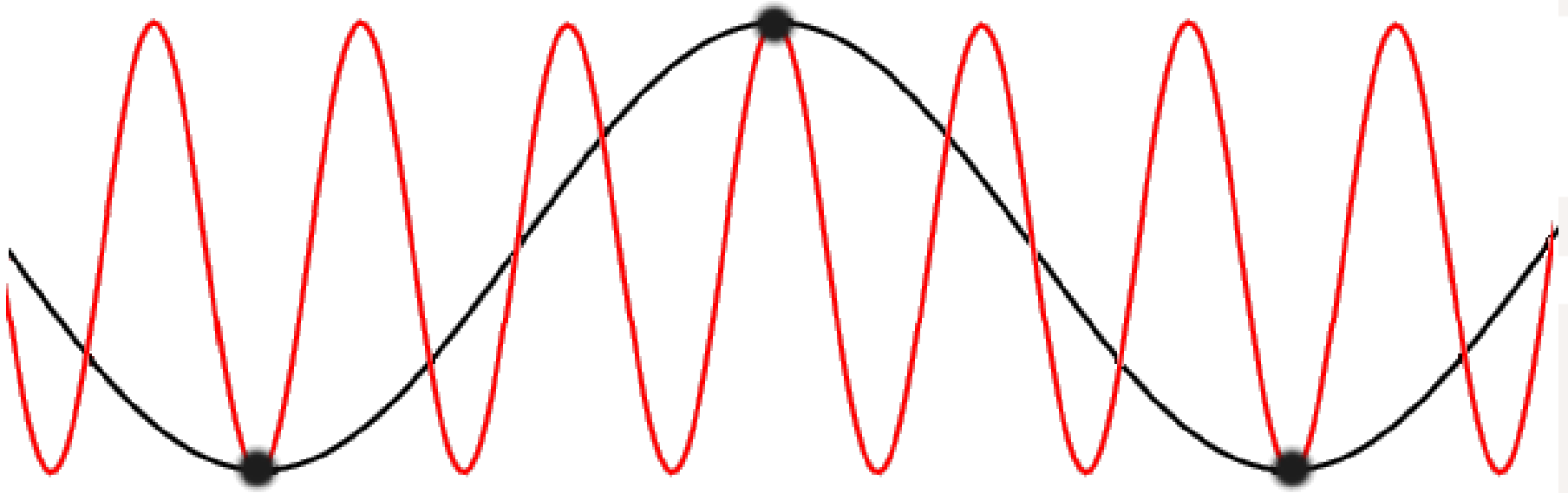
There are also second, third, *etc*., Brillouin zones, corresponding to a sequence of disjoint regions (all with the same volume) at increasing distances from the origin, but these are used less frequently

In general, the *n*-th Brillouin zone consists of the set of points that can be reached from the origin by crossing exactly *n* − 1 distinct Bragg planes



III. The First Brillouin Zone (BZ)

Why is the First BZ so important?



***k*-vectors exceeding the first Brillouin zone (red) do not carry any more information than their counterparts (black) in the first Brillouin zone**

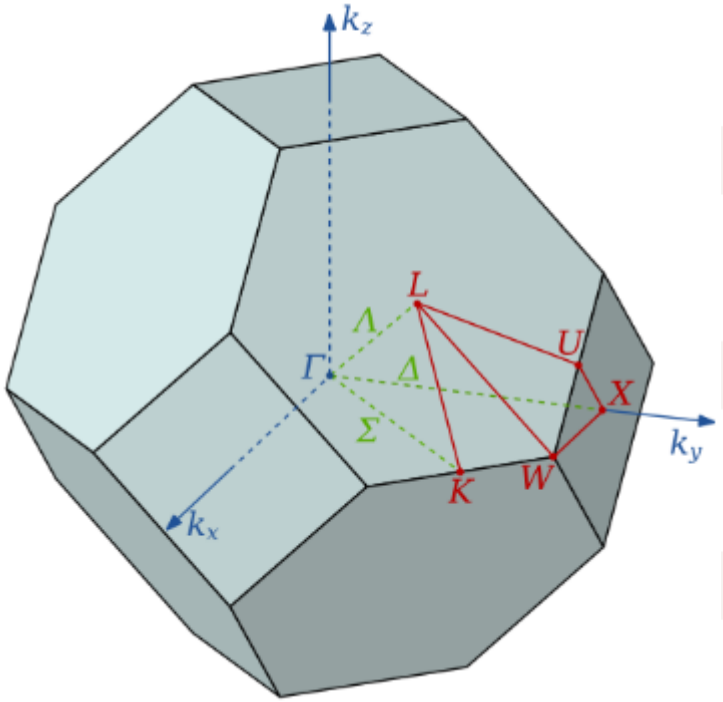
k* at the Brillouin zone edge is the spatial Nyquist frequency of waves in the lattice, because it corresponds to a half-wavelength equal to the inter-atomic lattice spacing *a

https://en.wikipedia.org/wiki/Brillouin_zone

IV. High-Symmetry Points and Irreducible BZ

Several points of high symmetry are of special interest – these are called critical points

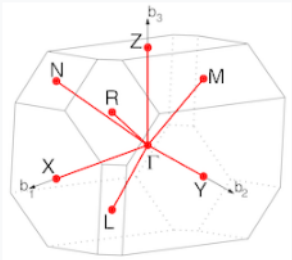
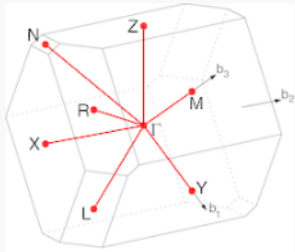
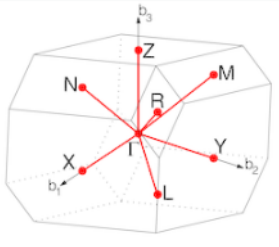
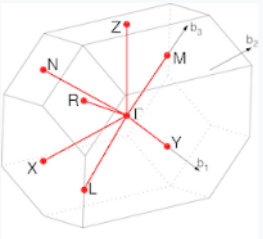
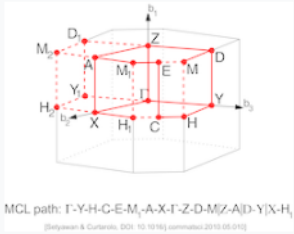
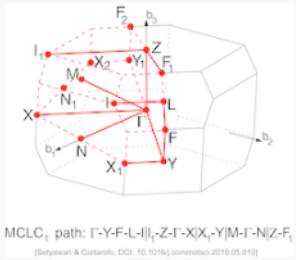
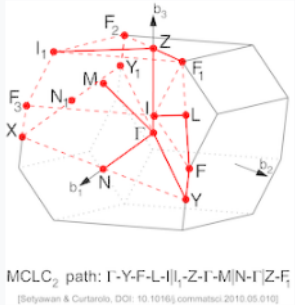
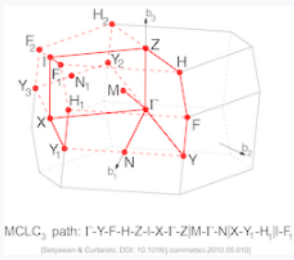
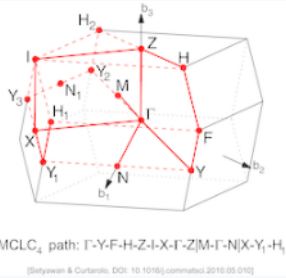
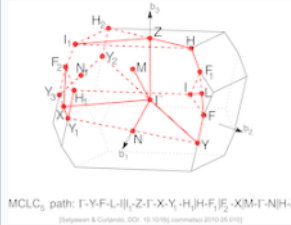
Symbol	Description
Γ	Center of the Brillouin zone
Simple cube	
M	Center of an edge
R	Corner point
X	Center of a face
Face-centered cubic	
K	Middle of an edge joining two hexagonal faces
L	Center of a hexagonal face
U	Middle of an edge joining a hexagonal and a square face
W	Corner point
X	Center of a square face
Body-centered cubic	
H	Corner point joining four edges
N	Center of a face
P	Corner point joining three edges
Hexagonal	
A	Center of a hexagonal face
H	Corner point
K	Middle of an edge joining two rectangular faces
L	Middle of an edge joining a hexagonal and a rectangular face
M	Center of a rectangular face



First Brillouin zone of FCC lattice

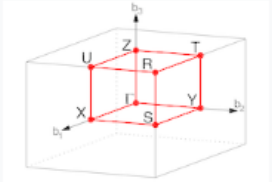
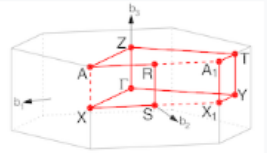
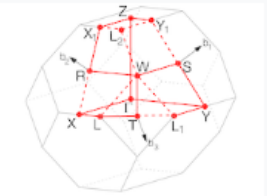
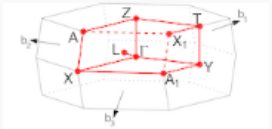
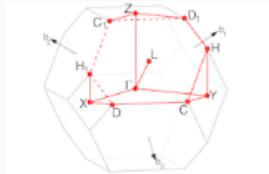
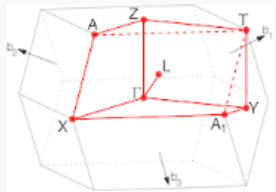
https://en.wikipedia.org/wiki/Brillouin_zone

IV. High-Symmetry Points and Irreducible BZ

Lattice system[4]	Bravais lattice (Abbreviation)					
Triclinic	Primitive triclinic (TRI)	Triclinic Lattice type 1a (TRI1a)  TRI_{1a} path: X-Γ-Y|L-Γ-Z|N-Γ-M|R-Γ	Triclinic Lattice type 1b (TRI1b)  TRI_{1b} path: X-Γ-Y|L-Γ-Z|N-Γ-M|R-Γ	Triclinic Lattice type 2a (TRI2a)  TRI_{2a} path: X-Γ-Y|L-Γ-Z|N-Γ-M|R-Γ	Triclinic Lattice type 2b (TRI2b)  TRI_{2b} path: X-Γ-Y|L-Γ-Z|N-Γ-M|R-Γ	
Monoclinic	Primitive monoclinic (MCL)	Monoclinic Lattice (MCL)  MCL path: Γ-Y-H-C-E-M_1-A-X-Γ-Z-D-M|Z-A|D-Y|X-H_1				
	Base-centered monoclinic (MCLC)	Base-centered monoclinic lattice type 1 (MCLC1)  $MCLC_1$ path: Γ-Y-F-L-I||I_1-Z-Γ-X|X$_1$-Y|M-Γ-N|Z-F_1	Base-centered monoclinic lattice type 2 (MCLC2)  $MCLC_2$ path: Γ-Y-F-L-I||I_1-Z-Γ-M|N-Γ|Z-F_1	Base-centered monoclinic lattice type 3 (MCLC3)  $MCLC_3$ path: Γ-Y-F-H-Z-I-X-Γ-Z|M-Γ-N|X-Y_1-H_1|I-F_1	Base-centered monoclinic lattice type 4 (MCLC4)  $MCLC_4$ path: Γ-Y-F-H-Z-I-X-Γ-Z|M-Γ-N|X-Y_1-H_1	Base-centered monoclinic lattice type 5 (MCLC5)  $MCLC_5$ path: Γ-Y-F-L-I||I_1-Z-Γ-X-Y_1-H_1|H-F_1|F_2-X|M-Γ-N|H-Z

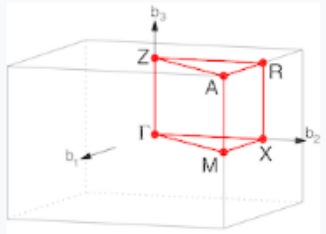
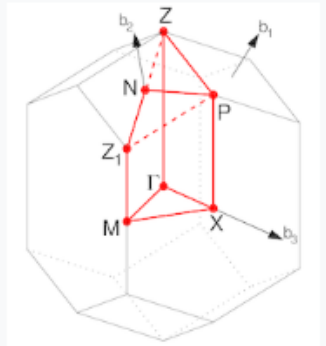
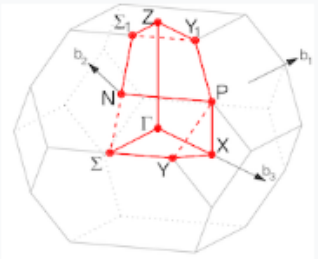
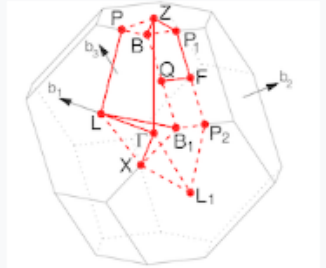
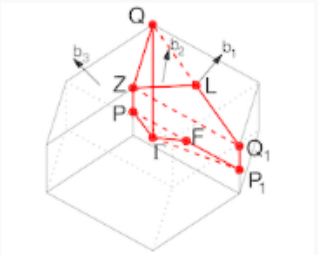
https://en.wikipedia.org/wiki/Brillouin_zone

IV. High-Symmetry Points and Irreducible BZ

Orthorhombic	Primitive orthorhombic (ORC)	Simple orthorhombic lattice (ORC)  ORC path: Γ-X-S-Y-Γ-Z-U-R-T-Z|Y-T|U-X|S-R		
	Base-centered orthorhombic (ORCC)	Base-centered orthorhombic lattice (ORCC)  ORCC path: Γ-X-S-R-A-Z-Γ-Y-X_1-A_1-T-Y|Z-T		
	Body-centered orthorhombic (ORCI)	Body-centered orthorhombic lattice (ORCI)  ORCI path: Γ-X-L-T-W-R-X_1-Z-Γ-Y-S-W|L_1-Y|Y_1-Z		
	Face-centered orthorhombic (ORCF)	Face-centered orthorhombic lattice type 1 (ORCF1)  $ORCF_1$ path: Γ-Y-T-Z-Γ-X-A_1-Y|T-X_1|X-A-Z|L-Γ	Face-centered orthorhombic lattice type 2 (ORCF2)  $ORCF_2$ path: Γ-Y-C-D-X-Γ-Z-D_1-H-C|C_1-Z|X-H|H-Y|L-Γ	Face-centered orthorhombic lattice type 3 (ORCF3)  $ORCF_3$ path: Γ-Y-T-Z-Γ-X-A_1-Y|X-A-Z|L-Γ

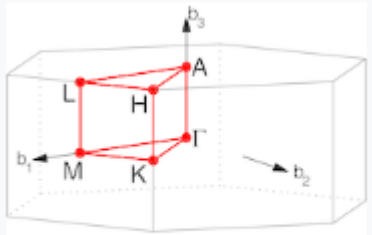
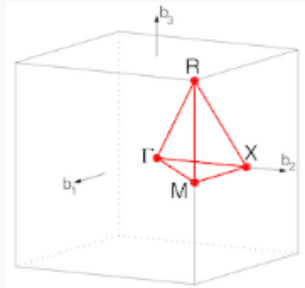
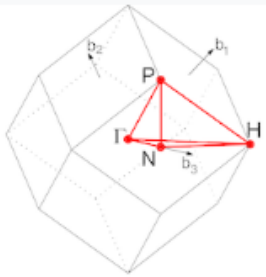
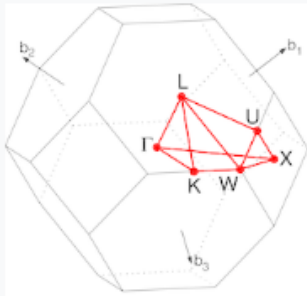
https://en.wikipedia.org/wiki/Brillouin_zone

IV. High-Symmetry Points and Irreducible BZ

Tetragonal	Primitive tetragonal (TET)	Simple tetragonal lattice (TET)  TET path: Γ-X-M-Γ-Z-R-A-Z|X-R|M-A	
	Body-centered Tetragonal (BCT)	Body-centered tetragonal lattice type 1 (BCT1)  BCT_1 path: Γ-X-M-Γ-Z-P-N-Z_1-M|X-P	Body-centered tetragonal lattice type 2 (BCT2)  BCT_2 Path: Γ-X-Y-Σ-Γ-Z-Σ_1-N-P-Y_1-Z|X-P
Rhombohedral	Primitive rhombohederal (RHL)	Rhombohedral lattice type 1 (RHL1)  RHL_1 path: Γ-L-B_1|B-Z-Γ-X|Q-F-P_1-Z|L-P	Rhombohedral lattice type 2 (RHL2)  RHL_2 path: Γ-P-Z-Q-Γ-F-P_1-Q_1-L-Z

https://en.wikipedia.org/wiki/Brillouin_zone

IV. High-Symmetry Points and Irreducible BZ

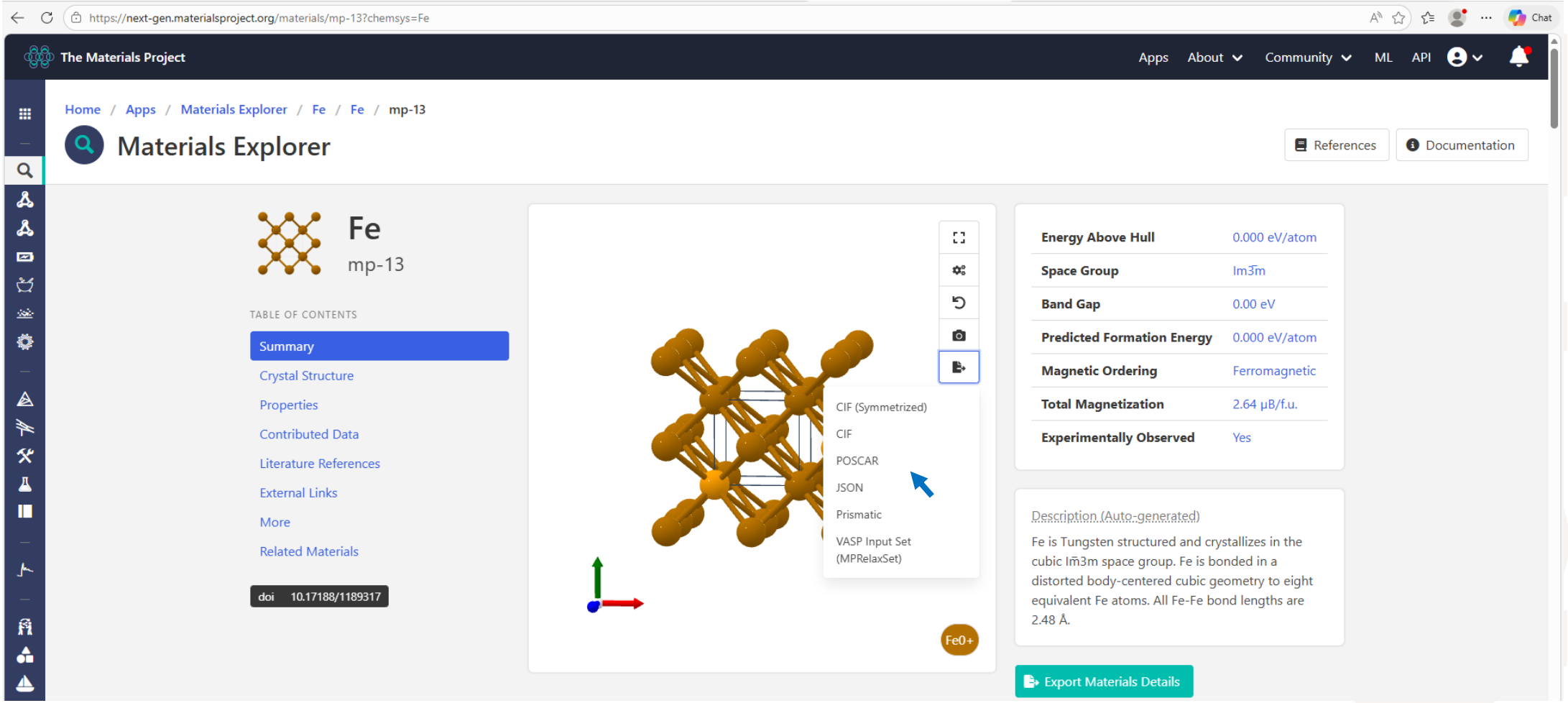
Hexagonal	Primitive hexagonal (HEX)	Hexagonal lattice (HEX)  HEX path: Γ-M-K-Γ-A-L-H-A|L-M|K-H
Cubic	Primitive cubic (CUB)	Simple cubic lattice (CUB)  CUB path: Γ-X-M-Γ-R-X|M-R
	Body-centered cubic (BCC)	Body-centered cubic lattice (BCC)  BCC path: Γ-H-N-Γ-P-H|P-N
	Face-centered cubic (FCC)	Face-centered cubic lattice (FCC)  FCC path: Γ-X-W-K-Γ-L-U-W-L-K|U-X

https://en.wikipedia.org/wiki/Brillouin_zone

IV. High-Symmetry Points and Irreducible BZ

Step One

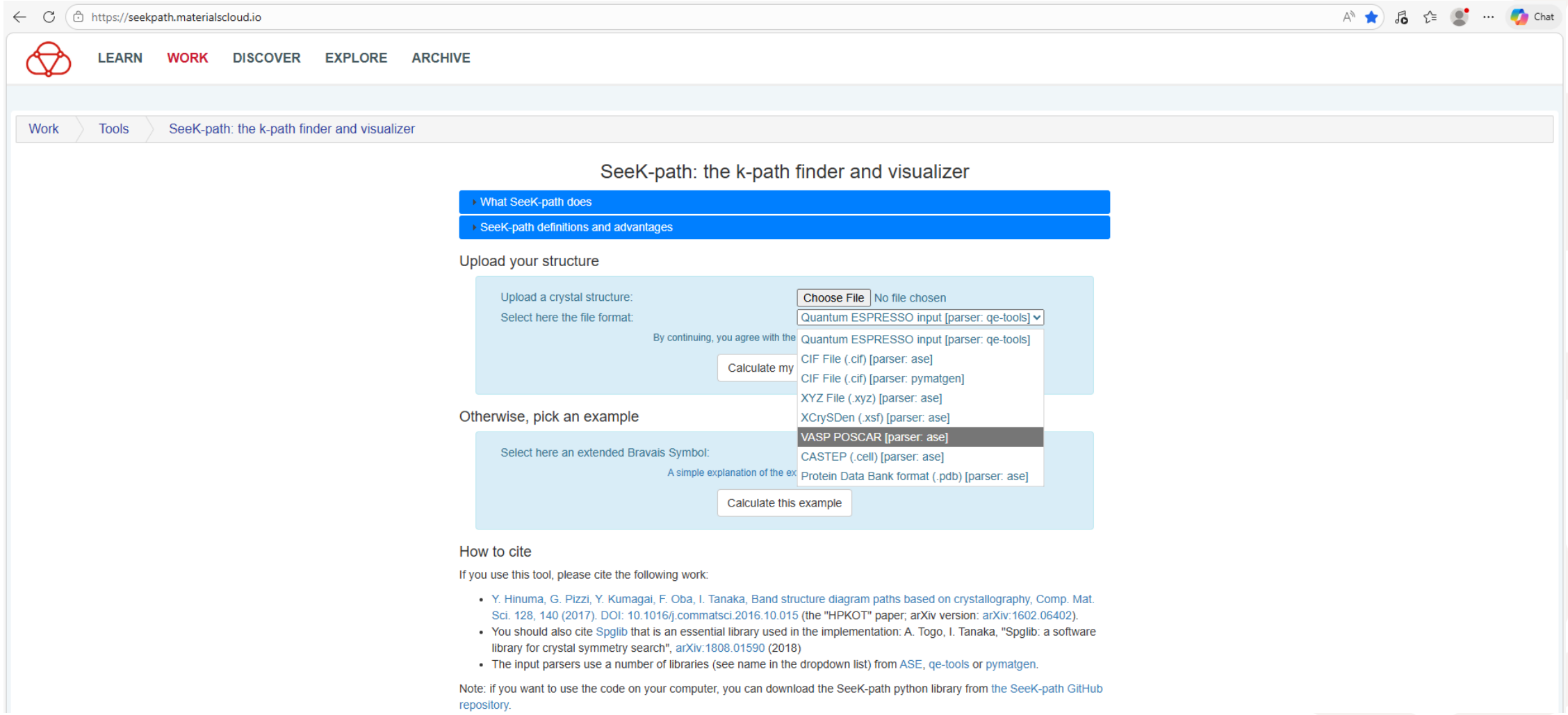
Export structure



IV. High-Symmetry Points and Irreducible BZ

Step Two

Upload structure



Xi'an Jiaotong-Liverpool University
西交利物浦大学

https://seekpath.materialscloud.io/

IV. High-Symmetry Points and Irreducible BZ

https://seekpath.materialscloud.io/compute/process_structure/

LEARN WORK DISCOVER EXPLORE ARCHIVE More

Work > Tools > SeeK-path: the k-path finder and visualizer

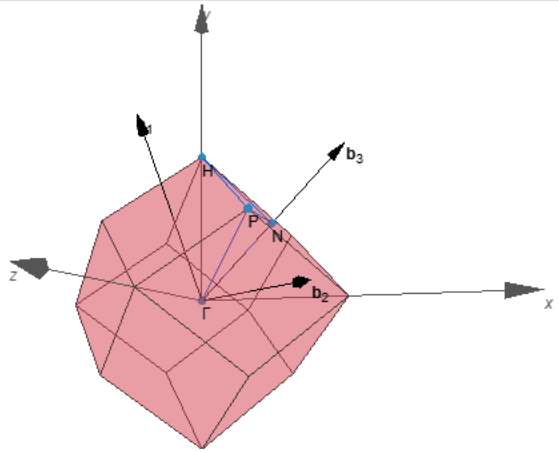
SeeK-path: the *k*-path finder and visualizer

▸ Show parsed input coordinates (please double-check here if the parser worked properly)

Jump directly to downloadable coordinates

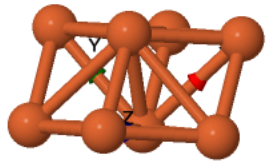
Brillouin zone (go to coordinates)

Drag to rotate, scroll to zoom, double-click to enable/disable interaction



Primitive structure (go to coordinates)

Drag to rotate, scroll to zoom, double-click to enable/disable interaction



Supercell: 2 2 2 UPDATE RESET 2x2x2 CELL

Camera: x y z

Axes: xyz axes

bonds [?] atom labels [?]

packed cell [?] space-filling [?]

rotation

IV. High-Symmetry Points and Irreducible BZ

https://seekpath.materialscloud.io/compute/process_structure/

Reciprocal space and Brillouin-zone information

Reciprocal cell vectors (1/Å)

b	x	y	z
b_1	-0.0000000000	2.1945886838	2.1945886838
b_2	2.1945886838	0.0000000000	2.1945886838
b_3	2.1945886838	2.1945886838	0.0000000000

Suggested path

Γ—H—N—Γ—P—H|P—N

High-symmetry points (scaled units)

Coordinates of *k* points

Label	k_1	k_2	k_3
Γ	0.0000000000	0.0000000000	0.0000000000
H	0.5000000000	-0.5000000000	0.5000000000
N	0.0000000000	0.0000000000	0.5000000000
P	0.2500000000	0.2500000000	0.2500000000

High-symmetry points (1/Å)

Label	k_x	k_y	k_z
Γ	0.0000000000	0.0000000000	0.0000000000
H	0.0000000000	2.1945886838	0.0000000000
N	1.0972943419	1.0972943419	0.0000000000
P	1.0972943419	1.0972943419	1.0972943419

Structure information (primitive cell)

Crystal structure information

Bravais lattice type: cI

Extended Bravais lattice symbol: cI1 (with inversion symmetry)

Spacegroup: Im-3m (number 229)

Primitive cell vectors (Å)

v	x	y	z
v_1	-1.4315177495	1.4315177495	1.4315177495
v_2	1.4315177495	-1.4315177495	1.4315177495
v_3	1.4315177495	1.4315177495	-1.4315177495

Atom coordinates (scaled)

Element	r_1	r_2	r_3
Fe	0.0000000000	0.0000000000	0.0000000000

Atom coordinates (Cartesian, Å)

Element	x	y	z
Fe	0.0000000000	0.0000000000	0.0000000000

How to cite and acknowledgements

If you use this tool, please cite the following work:

- Y. Hinuma, G. Pizzi, Y. Kumagai, F. Oba, I. Tanaka, Band structure diagram paths based on crystallography, Comp. Mat. Sci. 128, 140 (2017). DOI: 10.1016/j.commatsci.2016.10.015 (the "HPKOT" paper; arXiv version: arXiv:1602.06402).
- You should also cite Spglib that is an essential library used in the implementation: A. Togo, I. Tanaka, "Spglib: a software library for crystal symmetry search", arXiv:1808.01590 (2018)
- The input parsers use a number of libraries (see name in the dropdown list) from ASE, qe-tools or pymatgen.

Important note

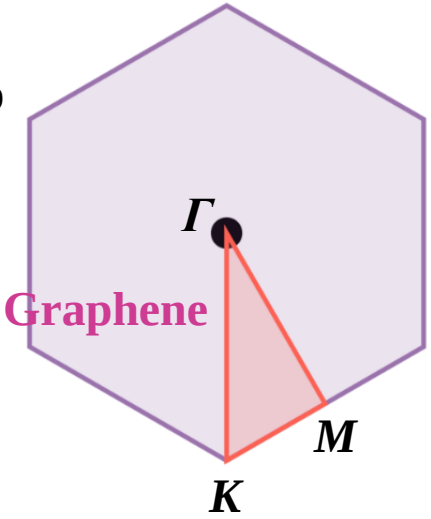
The structure visualized above is *not* the structure you passed in input, but its primitive cell. Moreover, the primitive cell has been standardized (using the conventions of spglib).
If you want to run a new simulation, and plan (now or later) to use the suggested path provided here, please use the primitive cell above.

IV. High-Symmetry Points and Irreducible BZ

Due to point-group symmetry (rotations/reflections), one does not need to calculate the whole 1st BZ

Only a tiny pie slice is enough

The smallest wedge of the First BZ that cannot be further reduced by any point-group symmetry operation of the crystal



The Irreducible BZ for Graphene

Crystals possess inherent symmetries (rotations, reflections, inversions). If the real-space crystal lattice is invariant under a symmetry operation S, its Reciprocal Lattice and Hamiltonian are also invariant $E_n(\hat{S}k) = E_n(k)$

Irreducible BZ usually covers all important information of the system; such as band gap, berry curvature, spin feature, *etc*.

https://en.wikipedia.org/wiki/Brillouin_zone



That's all for this session

Thank you for your attention and great participation